Relation Between the HDL Apoproteins and AI Isoproteins in Subjects With the AI_{Milano} Abnormality

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The apoprotein AI_{Milano} (AI_M) is the first described molecular abnormality of human apolipoproteins. In order to achieve a better understanding of the biochemical role of high density lipoproteins (HDL) and of their subcomponents, nine members of the AI_M family were studied. Five showed the characteristic biochemical features of the AI_M abnormality. All were hypertriglyceridemic; their plasma and very low density lipoprotein (VLDL) triglycerides had a significant negative correlation with the HDL cholesterol (HDL-C) levels ($r = -0.961$, $p < 0.01$ and $r = -0.939$, $p < 0.05$, respectively). Compositional studies of HDL in these subjects revealed a marked triglyceride (TG) enrichment, with reduced cholesterol content. HDL-C and/or phospholipid (PL) levels correlated significantly with the AI_M apoprotein in HDL ($r = 0.995$, $p < 0.001$ for HDL-C and $r = 0.994$, $p < 0.001$ for HDL-PL) but not with the other HDL apoproteins, suggesting that only apo AI_M in monomeric form, can bind lipids. Isoelectric focusing studies on the isolated AI_M isoproteins from the five affected subjects revealed an isoprotein distribution markedly different from normal AI. A possible structural role for isoprotein AI_1 is suggested by the strong correlation between HDL content of this isoprotein and HDL-C and HDL-PL levels ($r = 0.965$ and $r = 0.990$ respectively, both $p < 0.001$). AI_4 in AI_M was nearly doubled, as compared to normal AI. The reported results may be of help for a better understanding of the role of the major HDL apoprotein and of its isoproteins in lipid binding and lipoprotein metabolism.

AN APOPROTEIN AI (apo AI) variant, defined as AI_{Milano} (AI_M), characterized by cysteine content and disulphide bridge formation[1,2] has been detected in the high density lipoprotein (HDL) fraction of an Italian male subject. Major findings in this proband were markedly decreased HDL cholesterol (HDL-C) levels and elevated plasma triglycerides. The clinical examination was unremarkable, i.e., liver and ocular dysfunctions, described for other familial diseases associated with very low HDL-C levels,[3,4,5] were not detected. Vascular disease, generally believed to be related to decreased plasma HDL-C concentrations,[6,7] was found to be absent both in the proband and his family. Lipoprotein lipase (LPL) and lecithin-cholesterol acyl transferase (LCAT) activities were within normal limits. The HDL particles were enlarged and triglyceride (TG) enriched.[1] The abnormal apo AI_M forms complexes with apo AII (AI_M-AII) as well as dimers (AI_M-AI_M). Both the complex and the dimer are sensitive to treatment with thiols, breaking down into the respective subunits.

The availability of several other family members showing the typical signs of the abnormality, allowed the unique opportunity of examining in detail some of the traits of the disease, i.e., hypertriglyceridemia and decreased HDL-C levels. In particular, the HDL-C levels being distributed over a relatively wide range, a detailed analysis of lipid and apoprotein composition, with particular attention to the isoprotein forms of apo AI, was made possible.

MATERIALS AND METHODS

Patients and Controls

Studies were carried out in 7 members of the same family (Fig. 1). General characteristics of the proband (II-2, indicated by an arrow in the pedigree) and of his children were described in a previous report.[1] The age range of the patients was between 12 and 78 years; they were normotensive (SBP < 145) with regular heart rate at the time of investigation. In none could hepato- or splenomegaly be detected. The general physical examination was unremarkable. Body weights were normal or slightly elevated (90%–115% of the ideal BW). Laboratory tests were indicative of normal thyroid, liver and renal functions. ECG's at rest were normal; eye examination revealed normal grounds without corneal opacities.

One member of the family (II-1) reportedly died at age 26 due to complications of an abdominal infection. No biochemical data are available for this subject. The proband's mother (I-2) died recently following a stroke. As pointed out below, she had none of biochemical abnormalities found in the affected members of the family.

Ten normolipidemic subjects (age 21–54) acted as controls for the isoelectric focusing (IEF) studies. Lipid and lipoprotein levels were compared with data from a control population. Statistical analyses were carried out with the Student's t test and r correlation coefficient.

Lipoprotein Fractionation

Blood collected after an overnight fast, was anticoagulated with EDTA-Na_2 (1 mg/ml) and kept on ice; plasma was obtained by centrifuging at 900xg for 20'. It was then subjected to preparative ultracentrifugation in a Beckman L5-50 (Beckman, Palo Alto, USA) ultracentrifuge, by the use of a 50 Ti rotor at different salt densities. Density adjustments were made with solid KBr, according to standard techniques.[8] Very low and low density lipoproteins (VLDL and LDL) were recovered following centrifugation at 40,000 rpm respectively for 20 and 24 hr at 15°C. For the isolation

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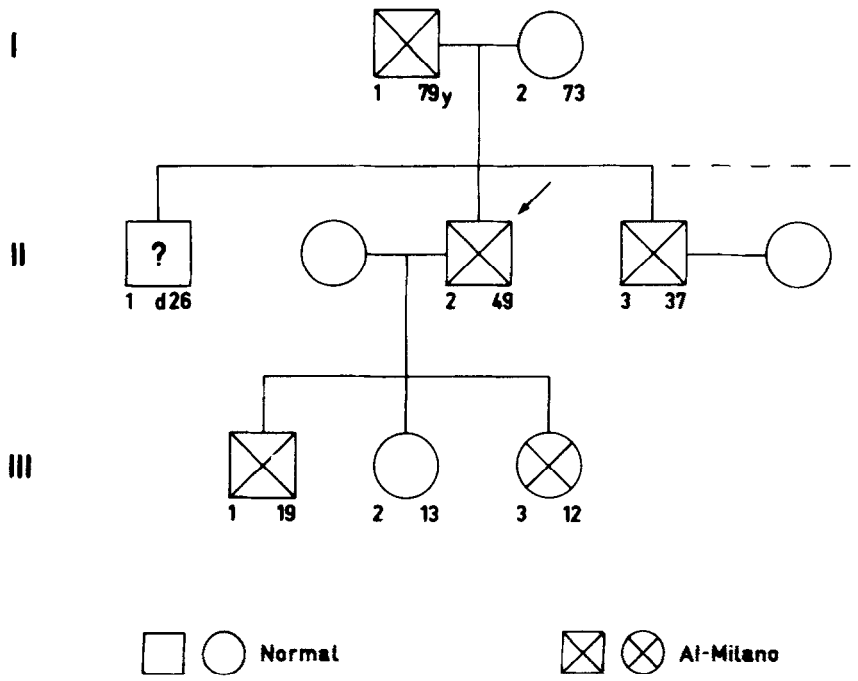


Fig. 1. Pedigree of the family members tested for the apo AI_{Milano} abnormality. Subject II-1 reportedly died at age 26 (biochemical data are not available). The proband is indicated by an arrow.

of HDL, density was raised to 1.25 and centrifugations carried out for 40 hr at 15°C. The supernatant was recovered and washed once by recentrifugation in salt solutions of the appropriate density. All fractions, except for VLDL, were dialyzed extensively against 0.15 M NaCl, 10^{-3} M EDTA, pH 7.4 at 4°C.

For reduction and alkylation, nondelipidated HDL (approximately 3 mg of protein in 3 ml of 0.2 M phosphate buffer, pH 8.0) were treated with 30 μl of 2-mercaptoethanol for 4 hr at room temperature. 90 mg of iodoacetamide were then added and the reaction continued for 30′ at 0°C in the dark. Excess reagent was then removed by dialysis against 0.15 M NaCl, 10^{-3} M EDTA, pH 7.4 at 4°C.

Lipoprotein Characterization

Total cholesterol (TC) and TG determinations were made on plasma and isolated lipoproteins using standard automated techniques.[9,10] The ratio of free to esterified cholesterol (FC/EC) was determined by an enzymatic method.[11] Protein analysis was by the method of Lowry et al.[12] and phospholipids (PL) by the method of Rouser et al.[13] HDL-cholesterol levels were also determined after selective precipitation of VLDL and LDL according to Kostner.[14] Aliquots of whole plasma and isolated lipoproteins were subjected to electrophoresis on agarose[15] and polyacrylamide gels.[16]

Preparation of Apoproteins

The lipoprotein fractions were delipidated by extraction with ethanol diethylether (3:1) at 4°C for 20 hr, followed by two other extractions of 4 hr each.[17] The precipitates were recovered after centrifugation at 3,000 rpm for 20′ and washed once with diethylether.

Electrophoresis and IEF of Apolipoproteins

Sodium dodecylsulphate (SDS) electrophoresis was performed on 0.1% gels by the method of Weber and Osborn,[18] the acrylamide concentration being 10% or 15%. In selected samples, the disulphide bonds were reduced by the addition of 5 μl of 2-mercaptoethanol to the solubilizing mixture. After staining of the gels in standard Coomassie Blue-methanol-acetic acid solution, apoprotein bands were identified from their molecular weights (MW), calculated from the relation of Rf vs log MW. Marker proteins were: bovine serum albumin (68,000 MW), egg albumin (43,000), pepsin (35,000), trypsin (23,000) and cytochrome C (11,700).

IEF was carried out by a modification of the method of Gidez et al.[19] Gels (0.5×8 cm) were prepared using 7.5% acrylamide and 8 M urea with 2% ampholites in the pH 4.0–6.0 range (Ampholine, LKB, Bromma, Sweden). Following a pre-run of 20′ at 100 V and 20′ at 200 V, the samples (50–100 μg of proteins in 100 μl of 8 M urea, 10% sucrose) were placed on top of the gels and focalized for 1 hr at 200 V and for 16–18 hr at 400 V at the constant temperature of 10°C, using the buffer system of Gidez.[19] The gels were fixed in 10% (w/v) trichloroacetic acid and stained with Coomassie Blue according to Malik and Berrie.[20] The pI of the separated isoproteins was determined by slicing an unloaded gel in correspondence with the protein bands and by shaking the gel slices in plastic vials containing 0.5 ml distilled water for 6 hr. The pH of the eluate was assumed as the apparent pI of the specific isoproteins.

Apoproteins were quantitated by densitometry of the SDS and IEF stained gels using a computerized Cellomatic 2A densitometer (CGA, Florence, Italy). The optical density recorded for each group of bands increased linearly with increasing amounts of loaded apoproteins up to 20 μg for each tested protein.

IEF Analysis of Isolated Apoproteins

After SDS gel electrophoresis (100–200 μg of proteins on 0.7×10 cm gels), the gels were shaken for 10–15′ in 20 ml of 4 M Na-acetate, thus precipitating the SDS unbound to the proteins.[21] The gels were then cut in correspondence with the transparent protein bands and the slices placed in plastic vials containing 1 ml of 8 M urea, 10% sucrose. IEF was carried out as previously described. In some cases, apoproteins were isolated by preparative SDS gel electrophoresis according to Stephens et al.[29] and analyzed by IEF; the results were comparable to those obtained by the described simplified procedure.

Table 1. Plasma and Lipoprotein Lipids in the AI_{Milano} Family

	Pedigree		I-1	II-2	II-3	III-1	III-3
Plasma							
	TC	(mg/dl)	160.2	231.4	275.5	190.3	183.9
	EC		96.8	160.8	189.9	135.0	131.3
	FC		63.4	70.6	85.6	55.3	52.6
	TG		215.0	318.8	228.3	272.6	180.6
	PL		185.0	280.0	180.5	125.3	125.6
VLDL							
	TC		23.4	40.8	26.5	26.6	11.5
	EC		12.0	21.1	15.2	11.4	5.8
	FC		11.4	19.7	11.3	15.2	5.7
	TG		124.3	206.0	120.8	189.0	102.0
	PL		62.0	74.0	62.5	57.5	24.6
	Prot.		10.5	38.1	13.2	24.9	15.8
LDL							
	TC		124.3	183.9	238.3	156.3	154.8
	EC		76.8	134.6	167.2	117.9	112.7
	FC		47.5	49.3	71.1	38.4	42.1
	TG		78.5	105.0	96.5	75.0	67.0
	PL		87.3	184.8	87.0	43.8	51.9
	Prot.		106.7	210.1	166.4	89.1	88.3
HDL*							
	TC		12.9 ± 0.4	6.3 ± 0.6	11.3 ± 0.6	7.3 ± 0.4	17.2 ± 0.8
	EC		8.3 ± 0.5	5.0 ± 0.1	7.9 ± 0.3	5.6 ± 0.1	12.5 ± 0.3
	FC		4.6 ± 0.1	1.3 ± 0.4	3.4 ± 0.3	1.7 ± 0.3	4.7 ± 0.4
	TG		12.2 ± 0.7	7.7 ± 0.5	11.0 ± 0.5	7.9 ± 0.6	10.5 ± 0.7
	PL		35.7 ± 1.7	21.2 ± 1.2	31.0 ± 1.6	24.0 ± 1.1	49.1 ± 1.8
	Prot.		48.1 ± 1.4	27.0 ± 1.2	45.4 ± 1.7	33.5 ± 1.4	65.3 ± 2.1

*$\bar{X}$ ± SD of three determinations.

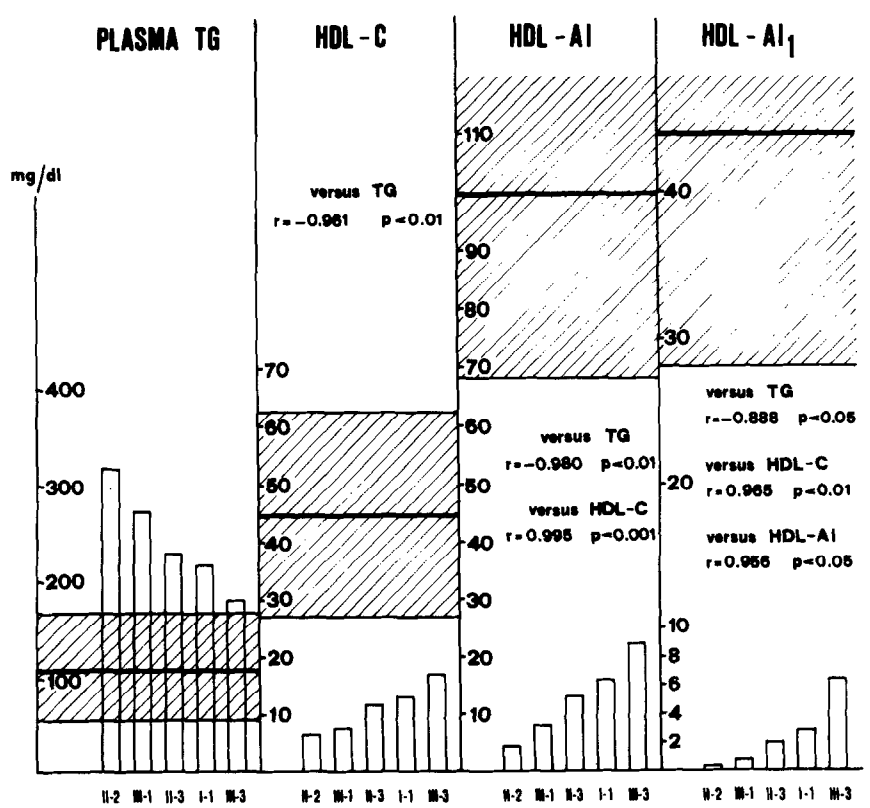


Fig. 2. Lipid and lipoprotein parameters and correlations among them in the five AI_M affected subjects. The mean and 95th percentile of the values from a control populations are indicated by the dashed areas.

Table 2. HDL Composition (% of mass) in the AI_{Milano} Family

Pedigree	TC	EC	FC	TG	PL	Prot.
I-1	11.8	7.6	4.2	11.2	32.8	44.2
II-2	10.1	8.0	2.1	12.4	34.1	43.4
II-3	11.4	8.0	3.4	11.1	31.4	46.0
III-1	10.0	7.7	2.3	10.9	33.0	46.1
III-3	10.8	7.8	3.0	6.6	30.8	41.0
$\bar{X} \pm$ SD: Patients	10.8 ± 0.8*	7.8 ± 0.2*	3.0 ± 0.9	10.4 ± 2.2*	32.4 ± 1.3	44.1 ± 2.1
$\bar{X} \pm$ SD: Controls (n = 10)	18.0 ± 2.1	14.5 ± 1.4	3.5 ± 0.7	3.9 ± 1.4	31.3 ± 3.2	46.8 ± 3.1

*$p < 0.001$ vs controls.

RESULTS

Plasma Lipids and Lipoproteins

The five affected members of this family (Fig. 1) constantly exhibited a variable hypertriglyceridemia. The lipoprotein pattern on agarose and polyacrylamide gels had a typical type IV profile for four patients (I-1, II-2, III-1, III-3), the fifth having a type IIb lipoprotein pattern. Compositional studies of plasma lipoproteins revealed a TG elevation in all lipoprotein classes (Table 1). The unaffected members of the family showed normal plasma lipid levels and lipoprotein distribution. Markedly reduced plasma HDL-C levels were also exhibited by all five affected members. This finding was confirmed by ultracentrifugation and selective precipitation. In this small group of subjects the HDL-C levels were significantly correlated with plasma TG ($r = -0.961$, $p < 0.01$) and with VLDL-TG levels ($r = -0.939$, $p < 0.05$) (Fig. 2).

The percentage composition of HDL, isolated by ultracentrifugation from the five affected subjects and from the control group is reported in Table 2. The abnormal HDL constantly exhibited reduced cholesterol content (particularly EC) with a marked TG enrichment. The SDS gel electrophoretic pattern of HDL-apoproteins of the affected family members revealed, as previously described for the proband,[2] the presence of three apoproteins not evident in the HDL of normal subjects. These apoproteins were identified as apo AI_M (MW 28,000), AI_M-AII (MW 35,000) and AI_M-AI_M (MW 55,000). The relative distribution in the five subjects was significantly different from one another (Table 3). From the data of Table 3, the concentrations of the individual HDL-apoproteins in the five subjects could be calculated: a significant correlation (Fig. 2) was found between the AI_M content of HDL (I-1 = 15.8, II-2 = 4.6, II-3 = 13.0, III-1 = 8.0, III-3 = 21.9 mg/dl) and the levels of HDL cholesterol ($r = 0.995$, $p < 0.001$) and HDL phospholipids ($r = 0.994$, $p < 0.001$).

IEF Studies

The apo HDL patterns of a normal subject and of the five family members with the AI_M abnormality after focusing on a narrow pH gradient (4.0–6.0) is shown in Fig. 3. The separation pattern, relative proportions and apparent pI of the major apo-HDL bands were reproducible between donors and experiments.

The difficulty in quantitating the relative distribution of the different bands in the AI region suggested that identification of their origin should be achieved by a combined technique of SDS electrophoretic separation and IEF. Well defined bands, corresponding to the AI_M monomer, dimer and AI_M-AII complex, were sliced from the SDS gels and subjected to IEF (Fig. 4). All abnormal apoproteins focused in the same region and contributed to the abnormal pattern (Fig. 5). Particularly AI_M apoproteins constantly showed five isoproteins with pI in the range 5.28–5.61 (AI_1, pH 5.61 ± 0.04; AI_2, pH 5.49 ± 0.05; AI_3, pH 5.42 ± 0.05; AI_4, pH 5.35 ± 0.04; AI_5, pH 5.28 ± 0.03). Five bands with a nearly identical pI were detected after IEF of a similarly isolated, normal apo-AI. Small variations in the apparent pI, noted in different experiments, were probably related to the variable width of the gel slice taken for pH measurements. The separa-

Table 3. HDL-Apoprotein Distribution (% of Total Protein†) in the AI_{Milano} Family

Pedigree	AI_M	AI_M-AII	AI_M-AI_M	AII	C	Others
I-1	32.9 ± 1.4	24.9 ± 1.3	16.1 ± 1.3	7.5 ± 1.3	5.5 ± 1.5	13.2 ± 2.4
II-2	17.1 ± 1.8	32.4 ± 3.5	10.2 ± 2.7	26.9 ± 2.4	6.6 ± 1.4	6.9 ± 1.7
II-3	28.7 ± 1.0	31.6 ± 1.7	14.3 ± 1.3	9.8 ± 0.4	6.2 ± 1.5	9.3 ± 1.3
III-1	24.7 ± 1.6	32.6 ± 1.2	7.0 ± 2.0	12.9 ± 0.8	12.4 ± 0.9	10.4 ± 2.2
III-3	33.6 ± 1.4	25.4 ± 1.3	11.8 ± 0.9	9.9 ± 0.6	5.8 ± 1.7	11.9 ± 3.3
Controls	62.7 ± 3.6*	—	—	16.5 ± 2.3	11.8 ± 1.0	9.0 ± 3.1

*normal apo-AI.

†mean ± SD of the results from 5 different gels.

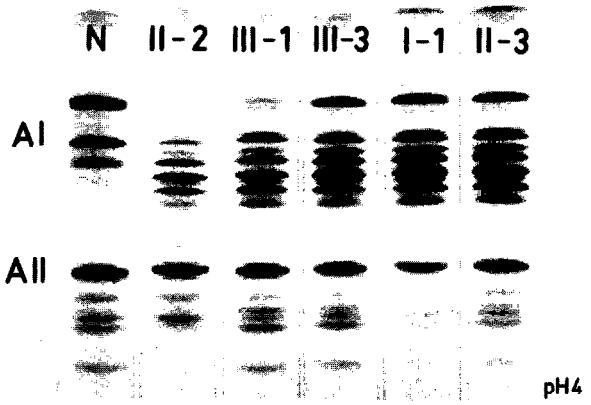


Fig. 3. Isoelectric focusing pattern (pH range 4–6) of isolated HDL from a normal subject and from the five AI_M affected subjects; each gel contains 70 μg of protein.

tion pattern and relative distribution of the five isoproteins were extremely reproducible.

The AI_M-AII apoproteins focused in a more acidic region than AI_M and showed more than 8 isoproteins. The AI_M-AI_M dimer consisted of 6 isoproteins with a narrow pI range, in the middle of the abnormal HDL pattern. It may be thus concluded that the most alkaline band in the AI region (AI_1, pH = 5.61 ± 0.04) is predominantly an isoprotein of the monomer AI_M.

Based on these findings, some important differences

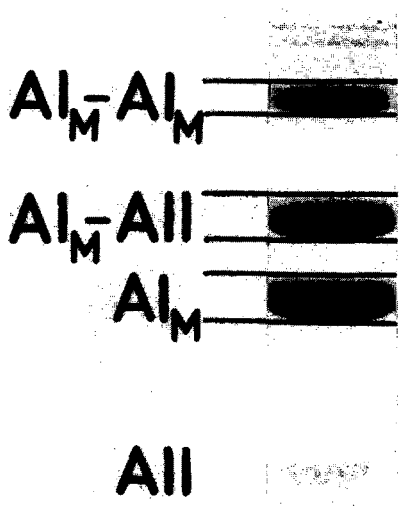


Fig. 4. Electrophoretic separation on SDS gel of HDL apoproteins from one of the studied family members. The gel regions corresponding to the apoproteins of different molecular weight are indicated. These were then sliced from the gels and subjected to IEF.

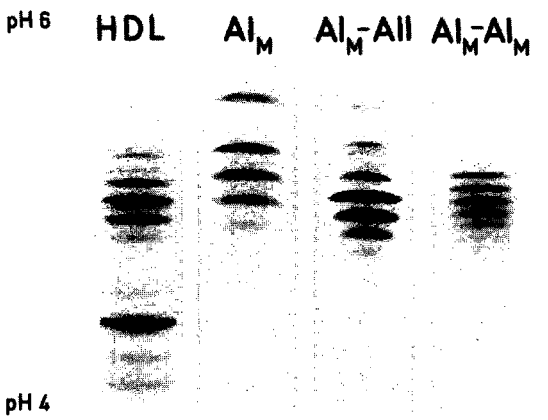


Fig. 5. Isoelectric focusing pattern (pH range 4.0–6.0) of total and individual HDL apoproteins isolated from the proband. All the anomalous apoproteins focus in the same region. The AI_M-AII complex and the AI_M-AI_M dimer consist of 8 and 6 isoproteins, respectively: AI_M shows five well distinct bands. For the experimental details see Methods.

could be detected between the isoprotein patterns of the different family members (Fig. 3). In fact, the most alkaline band of AI (AI_1) is clearly almost absent in the proband and in his son (II-2, III-1), being present in significant amounts in the other three affected members of the family. Statistically significant correlations could be found between the calculated amounts of HDL-AI_1 isoprotein (I-1 = 2.70; II-2 = 0.30; II-3 = 2.04; III-1 = 0.81; III-3 = 6.44 mg/dl) and the HDL-C and HDL-PL levels ($r = 0.965$ and $r = 0.990$, respectively, both $p < 0.001$) in the five affected subjects (Fig. 2).

The densitometric distribution of the isoproteins of AI_M (corresponding to those of normal apo AI) were also quantitated. The AI_M isoprotein ratios differed

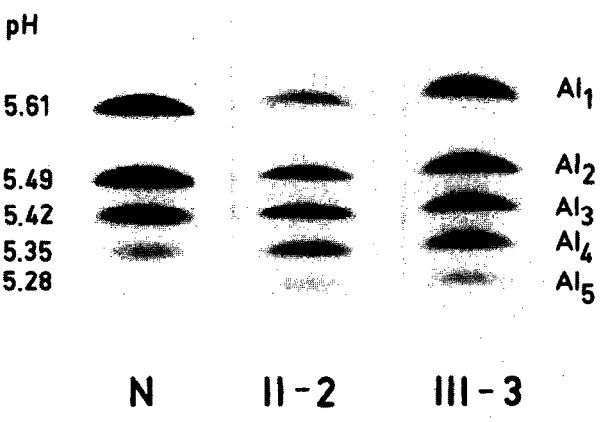


Fig. 6. Isoelectric focusing distribution of the five isoforms of apoprotein AI, isolated from a normal subject (N) and from two AI_M affected subjects. AI_M constantly shows five bands with the same pI of normal apo AI, but the AI_M isoprotein ratios differ significantly from N and from one another.

Table 4. Apo-AI Isoprotein Distribution in the AI_{Milano} Family*

Pedigree	AI_1	AI_2	AI_3	AI_4	AI_5
I-1	17.1 ± 0.8	25.1 ± 0.3	27.3 ± 2.0	20.3 ± 0.7	10.2 ± 0.9
II-2	6.6 ± 1.4	27.7 ± 1.1	36.4 ± 2.9	20.6 ± 0.5	8.7 ± 0.8
II-3	15.7 ± 1.0	27.7 ± 0.7	29.2 ± 0.9	18.7 ± 1.5	8.8 ± 0.4
III-1	10.1 ± 1.2	28.3 ± 1.8	36.6 ± 1.4	16.2 ± 1.8	8.8 ± 0.5
III-3	29.4 ± 0.1	27.5 ± 1.5	23.7 ± 0.4	14.8 ± 1.3	5.0 ± 0.4
mean ± SD	15.8 ± 8.7†	27.3 ± 1.2	30.6 ± 5.7‡	18.1 ± 2.5†	8.3 ± 1.9†
Controls	42.8 ± 4.3	26.4 ± 3.8	19.1 ± 3.3	9.8 ± 1.9	1.9 ± 0.5

*Percentage of total apo-AI (mean ± SD; results are from 3 different gels).
†$p < 0.002$. ‡$p < 0.01$ vs controls.

significantly from normal AI and from one another (Fig. 6). A doubling of mean AI_4 and a significant increase of AI_3 and AI_5, concomitant with the above reported dramatic reduction of AI_1, could be observed (Table 4).

DISCUSSION

The elucidation of the AI_{Milano} abnormality[1,2] represents an important contribution to the understanding of the physiology of lipid transport, and particularly of the metabolic role of HDL and of its components, especially in view of the possible protective effect of HDL against vascular disease.[23] This new abnormality differs from other familial disorders associated with low HDL levels. Tangier disease,[3] LCAT deficiency[4] and fish-eye disease[5] are all characterized by significant clinical symptoms associated with the biochemical abnormality, in striking contrast to the AI_M patients. Moreover, in Tangier disease an anomalous HDL particle containing only apo-AII has been detected,[24] whereas the HDL from the described patients contain all the known apoproteins, apo AI being substituted by a variant form (AI_M) containing cysteine and capable of forming complexes with apo-AII as well as dimers. These lipoprotein particles are enlarged and TG enriched.[1]

The observed disappearance of dimers and complexes following reduction and alkylation of HDL, with the consequent apparent increase of apo-AI and of apo AII on SDS gels, suggests that the abnormal larger apoproteins are at or near the lipoprotein surface. All disulphides would in this way be accessible to chemical modification. The anomalous apoproteins do not have, as appears from our data, a higher affinity for cholesterol than normal apo-HDL. This finding is of interest because the affected subjects, in spite of very low levels of HDL-C, do not show any clinical signs of atherosclerosis, nor is the disease significantly present in the family. An extreme longevity of some members of the kindred was also reported, in apparent contrast with the well recognized inverse correlation between levels of HDL-C and development of vascular disease.[23] The "longevity syndrome" has been, in fact, described in subjects with markedly elevated HDL-C concentrations.[25] Some other types of protective mechanism controlling tissue cholesterol removal should, therefore, be present in these patients. Awaiting data from turnover studies, the attention was focused on a better characterization of the abnormal HDL present in the AI_M affected subjects.

Nine members of the AI_M family were examined, distributed in three generations, with an extremely wide age range. Excluding the acquired relatives (the three spouses, all having normal lipid and lipoprotein parameters), five of the other six examined subjects had the characteristic biochemical features of the AI_M abnormality, with very low HDL levels and increased plasma TG. These parameters varied, however, very markedly from one subject to another. A very small population in which the disorder is present in forms of different severity could be thus studied, with a special interest for possible relationships among lipid, apoprotein and isoprotein parameters.

A widely described negative correlation was first detected in this small group of subjects, i.e., between HDL-C, plasma total and VLDL-TG levels.[26] Hypertriglyceridemia was not related, in these subjects, either to reduced LPL activity[27] or to a deficiency of LPL activators (unpublished data) as hypothesized for severe forms of the disease.[28]

Other correlations, more original and of possibly general interest, could be however established, due to the uniqueness of the apo-HDL pattern in these subjects. Most striking was the observed parallelism between the increase in the cholesterol content of HDL and that of apo AI_M and of its isoform AI_1 (Fig. 2). The highly significant correlation found between HDL-C and /or HDL-PL and the HDL content of the apo AI_M monomer, in the absence of a significant correlation with the other HDL apoproteins, suggests that only apo AI_M, in its monomeric form, can bind lipids. Also in the case of normal apo AI, which self-associates in aqueous solution, lipid binding occurs readily only in the monomeric form.[29] The large amount of apo AI_M involved in the formation of complexes and dimers in our patients, therefore being

probably unable to bind PL and cholesterol, can possibly explain the very low levels of plasma HDL-C. Lipid-binding studies for apo AI_M are, however, necessary to establish if the slightly different aminoacid composition can modify binding properties.

The extreme complexity of the apo-HDL IEF pattern in these subjects was partially explained by studies of isolated individual apoproteins. All the three abnormal apoproteins, clearly separated in SDS gels, focus in a narrow pH range, with partial overlap of the isoforms. AI_M-AI_M and AI_M-AII showed, respectively, six and eight different isoproteins, all focusing toward the acidic region of the gel area. In contrast, AI_M focused in five distinct bands, in perfect correspondence with the isoelectric points of normal AI isoproteins, as previously described in a preparative system.[30] The only minor difference was that normal apo AI shows only four isoprotein bands, whereas an extremely thin fifth band has been detected when examining normal AI as well as AI_M in analytical gels. Differences between the analytical systems can probably explain this small discrepancy. It has been well established that the apo AI polymorphism is not due to interactions of proteins with urea[31] or ampholytes.[32] All precautions were, however, taken to avoid any artifactual conditions and, moreover, the apo AI patterns were extremely reproducible between donors and experiments, in control subjects, as well as in the AI_M subjects.

The analysis of the apo AI isoprotein distribution showed a considerable constancy in the controls, whereas a significant variability in the percentage of AI_M isoproteins was observed in the family members. The very low apo AI_1 content and its variability in the affected subjects, highly correlated with the HDL-C and HDL-PL concentrations, suggests a possible structural role for isoprotein AI_1. Interestingly, a significant increase (almost a doubling) of the relative content of AI_4 was observed in the family. AI_4 has an apparently different aminoacid composition from other isoproteins, containing approximately one residue of isoleucine (Ile) per mole; moreover it appears to be a potent inhibitor of LCAT.[30]

The varieties of biochemical ratios characterizing the AI_M abnormality are probably all correlated among themselves. This presumption being valid, the described results would be of great interest, since the role of HDL and, particularly, of its major peptide AI in lipid metabolism could be better clarified. The possibility cannot, however, be dismissed that some or all lipoprotein abnormalities detected in the present studies are only related to an apoprotein AI variant (AI_M). This variant form maintains some properties of normal apo AI (MW, IEF pattern, antigenicity) and has a slightly different aminoacid composition.[2] Sequence studies are in progress to better define the structural abnormality; results from these studies will allow a more exact comparison between the AI_M and normal AI properties and functions.

REFERENCES

1. Franceschini G, Sirtori CR, Capurso A et al: The AI_{Milano} apoprotein: I Decreased high density lipoprotein cholesterol levels with significant lipoprotein modifications and without clinical atherosclerosis in an Italian family. J Clin Invest 66:892–900, 1980

2. Weisgraber KH, Franceschini G, Bersot TP et al: The AI_{Milano} apoprotein: II Isolation and characterization of a cysteine containing variant of the AI apoprotein from human high density lipoproteins. J Clin Invest 66:901–907, 1980

3. Fredrickson DS, Altrocchi PM, Avioli LV, et al: Tangier disease. Ann Intern Med 55:1016–1031, 1961

4. Gjone E, Norum KR, Glomset JA: Familial lecithin cholesterol acyltransferase deficiency, in Stanbury JB, Wyngaarden JB and Fredrickson DS (Eds): The Metabolic Basis of Inherited Disease. New York, McGraw-Hill, 1978, pp 589–603

5. Carlson LA, Philipson B: Fish-eye disease. A new familial condition with massive corneal opacities and dyslipoproteinemia. Lancet i:921–924, 1979

6. Miller GJ, Miller NE: Plasma high density lipoprotein concentration and development of ischemic heart disease. Lancet 1:16–19, 1975

7. Rössner S, Kjellin KG, Mettinger KL, et al: Normal serum cholesterol but low HDL-cholesterol concentration in young patients with ischemic cerebro-vascular disease. Lancet 1:577–581, 1978

8. Havel RJ, Eder HA, Bragdon JH: The distribution and chemical composition of ultracentrifugally separated lipoproteins in human serum. J Clin Invest 34:1345–1354, 1955

9. Block WD, Jarrett J, Levine JB: Use of a simple color reagent to improve the automated determination of serum cholesterol. In: Technicon Symposia on Automation in Analytical Chemistry, New York, Mediad Inc, pp 345–349, 1965

10. Noble RP, Campbell FM: Improved accuracy in automated fluorometric determination of plasma triglycerides. Clin Chem 16:166–170, 1971

11. Röschlau P, Bernt E, Gruber W: Enzymatische Bestimmung des Gesamt-cholesterins im Serum. Z Klin Chem Klin Biochem 12:403–407, 1974

12. Lowry OH, Rosebrough NJ, Farr AL, et al: Protein measurement with the Folin phenol reagent. J Biol Chem 193:265–275, 1951

13. Rouser G, Fleischer S, Yamamoto Y: Two dimensional thin layer chromatographic separations of polar lipids and determination of phospholipids by phosphorus analysis of spot. Lipids 5:434–496, 1970

14. Kostner GM: Enzymatic determination of cholesterol in high density lipoprotein fractions prepared by polyanion precipitation. Clin Chem 22:695–700, 1976

15. Noble RP: Electrophoretic separation of plasma lipoproteins in agarose gel. J Lipid Res 9:693–700, 1968

16. Naito HK, Woda M, Earhart LA, et al: Polyacrylamide gel electrophoresis as a screening procedure for serum lipoprotein abnormalities. Clin Chem 19:228–234, 1973

17. Brown WV, Levy RI, Fredrickson DS: Studies of the proteins

in human plasma very low density lipoproteins. J Biol Chem 244:5687–5694, 1969

18. Weber K, Osborn M: The reliability of molecular weight determinations by dodecyl sulphate-polyacrylamide gel electrophoresis. J Biol Chem 238:4406–4412, 1963

19. Gidez LI, Swaney JB, Murnane S: Analysis of rat serum apoprotein by isoelectric focusing. I Studies on the middle molecular weight subunits. J Lipid Res 18:59–68, 1977

20. Malik N, Berrie A: New stain fixative for proteins separated by gel isoelectric focusing based on Coomassie Brilliant Blue. Anal Biochem 43:173–176, 1972

21. Higgins RC, Dahmus ME: Rapid visualization of protein bands in preparative SDS-polyacrylamide gels. Anal Biochem 93:257–260, 1979

22. Stephen RE: High-resolution preparative SDS-polyacrylamide gel electrophoresis: fluorescent visualization and electrophoresis elution concentration of protein bands. Anal Biochem 65:369–379, 1975

23. Gordon T, Castelli WP, Hjortland MC, et al: High density lipoprotein as a protective factor against coronary heart disease. Am J Med 62:707–714, 1977

24. Assmann G, Herbert PM, Fredrickson DS, et al: Isolation and characterization of an abnormal high density lipoprotein in Tangier disease. J Clin Invest 60:242–252, 1977

25. Glueck CJ, Fallat RW, Millet F, et al: Familial hyperalphalipoproteinemia: studies in eighteen kindreds. Metabolism 24:1243–1265, 1975

26. Gotto AM Jr, Jackson RL: Plasma lipoproteins and atherosclerosis. Atherosclerosis Rev 3:231–242, 1978

27. Boberg J, Boberg M, Gross R, et al: Hepatic triglyceride and lipoprotein lipase activities of postheparin plasma in normals and hypertriglyceridemics. Uppsala J Med Sci 84:215–227, 1973

28. Kashyap ML, Srivastava LS, Chen CY, et al: Radioimmunoassay of human apolipoprotein CII. J Clin Invest 60:171–180, 1977

29. Vitello LB, Scanu AM: Studies on human serum high density lipoproteins. Self-association of apolipoprotein AI in aqueous solution. J Biol Chem 251:1131–1136, 1976

30. Nestruck AC, Suzue G, Marcel YL: Studies on the polymorphism of human apolipoprotein AI. Biochim Biophys Acta 617:110–121, 1980

31. Edelstein G, Lim CT, Scanu AM: On the subunit structure of the protein of human serum high density lipoprotein. I Study of its major polypeptide component (Sephadex fraction III). J Biol Chem 247:5842–5849, 1972

32. Scanu AM, Lim CT, Edelstein C: On the subunit structure of protein of human high density lipoprotein. II A study of Sephadex fraction IV. J Biol Chem 247:5850–5855, 1972